

Solving CSPs

Outline

I. Binary CSPs

II. Constraint Propagation

I. Binary CSP

A *unary constraint* restricts the value of a single variable.

$\langle (SA), SA \neq \textit{green} \rangle$

I. Binary CSP

A *unary constraint* restricts the value of a single variable.

$\langle (SA), SA \neq \textit{green} \rangle$

A *binary constraint* relates two variables.

$SA \neq WA$

I. Binary CSP

A *unary constraint* restricts the value of a single variable.

$\langle (SA), SA \neq \text{green} \rangle$

A *binary constraint* relates two variables.

$SA \neq WA$

A higher order constraint relate more variables.

Ternary constraint example: $\langle (X, Y, Z), X < Y < Z \text{ or } X > Y > Z \rangle$

I. Binary CSP

A *unary constraint* restricts the value of a single variable.

$\langle (SA), SA \neq \text{green} \rangle$

A *binary constraint* relates two variables.

$SA \neq WA$

A higher order constraint relate more variables.

Ternary constraint example: $\langle (X, Y, Z), X < Y < Z \text{ or } X > Y > Z \rangle$

A *global constraint* involves an arbitrary number of variables (but not necessarily all the variables).

I. Binary CSP

A *unary constraint* restricts the value of a single variable.

$\langle (SA), SA \neq \text{green} \rangle$

A *binary constraint* relates two variables.

$SA \neq WA$

A higher order constraint relate more variables.

Ternary constraint example: $\langle (X, Y, Z), X < Y < Z \text{ or } X > Y > Z \rangle$

A *global constraint* involves an arbitrary number of variables (but not necessarily all the variables).

Alldiff($v_1, \dots, v_k$): variables $v_1, \dots, v_k$ must have different values.

e.g., Sudoku (all variables in a row, column, or 3×3 box).

Cryptarithmic Puzzle

$$\begin{array}{r} T \quad W \quad O \\ + \quad T \quad W \quad O \\ \hline F \quad O \quad U \quad R \end{array}$$

Cryptarithmic Puzzle

$$\begin{array}{r} T \quad W \quad O \\ + \quad T \quad W \quad O \\ \hline F \quad O \quad U \quad R \end{array}$$

Addition constraints on the four columns:

$$\begin{aligned} O + O &= R + 10 \cdot C_1 \\ C_1 + W + W &= U + 10 \cdot C_2 \\ C_2 + T + T &= O + 10 \cdot C_3 \\ C_3 &= F \end{aligned}$$

Cryptarithmic Puzzle

$$\begin{array}{r}
 T \quad W \quad O \\
 + \quad T \quad W \quad O \\
 \hline
 F \quad O \quad U \quad R
 \end{array}$$

Addition constraints on the four columns:

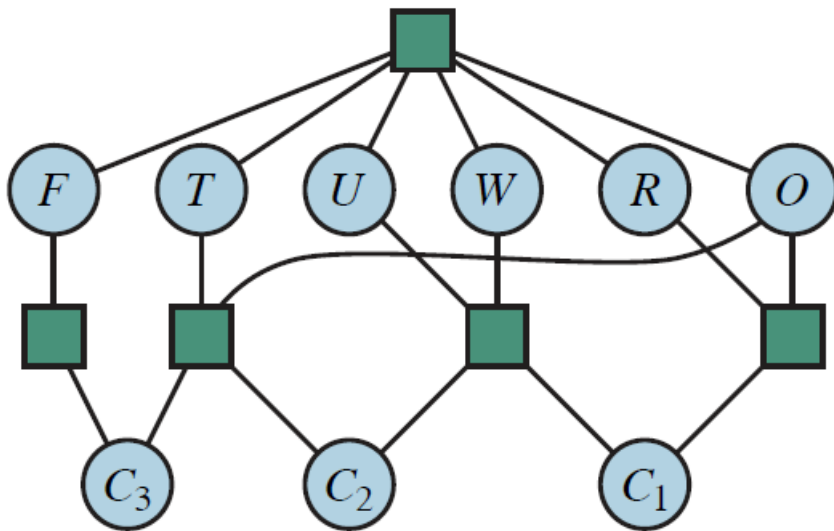
$$\begin{array}{l}
 O + O = R + 10 \cdot C_1 \\
 C_1 + W + W = U + 10 \cdot C_2 \\
 C_2 + T + T = O + 10 \cdot C_3 \\
 C_3 = F
 \end{array}
 \left. \begin{array}{l} C_1 \\ C_2 \\ C_3 \end{array} \right\} \begin{array}{l} \text{carry-overs into} \\ \text{the tens, hundreds,} \\ \text{and thousands,} \\ \text{Respectively.} \end{array}$$

Cryptarithmic Puzzle

$$\begin{array}{r}
 T \quad W \quad O \\
 + \quad T \quad W \quad O \\
 \hline
 F \quad O \quad U \quad R
 \end{array}$$

Addition constraints on the four columns:

$$\begin{array}{l}
 O + O = R + 10 \cdot C_1 \\
 C_1 + W + W = U + 10 \cdot C_2 \\
 C_2 + T + T = O + 10 \cdot C_3 \\
 C_3 = F
 \end{array}
 \begin{array}{l}
 \boxed{C_1} \\
 \boxed{C_2} \\
 \boxed{C_3}
 \end{array}
 \begin{array}{l}
 \text{carry-overs into} \\
 \text{the tens, hundreds,} \\
 \text{and thousands,} \\
 \text{Respectively.}
 \end{array}$$



Constraint hypergraph

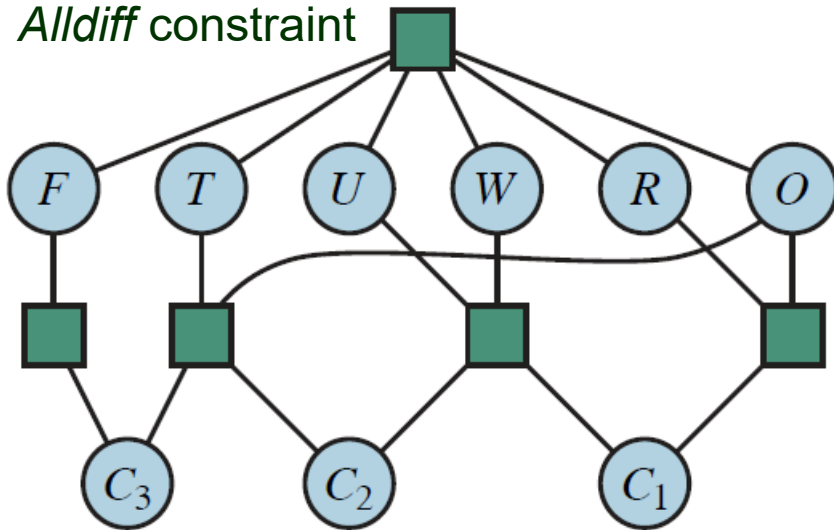
Cryptarithmic Puzzle

$$\begin{array}{r}
 T \quad W \quad O \\
 + \quad T \quad W \quad O \\
 \hline
 F \quad O \quad U \quad R
 \end{array}$$

Addition constraints on the four columns:

$$\begin{array}{l}
 O + O = R + 10 \cdot C_1 \\
 C_1 + W + W = U + 10 \cdot C_2 \\
 C_2 + T + T = O + 10 \cdot C_3 \\
 C_3 = F
 \end{array}
 \begin{array}{l}
 \boxed{C_1} \\
 \boxed{C_2} \\
 \boxed{C_3}
 \end{array}
 \begin{array}{l}
 \text{carry-overs into} \\
 \text{the tens, hundreds,} \\
 \text{and thousands,} \\
 \text{Respectively.}
 \end{array}$$

Alldiff constraint



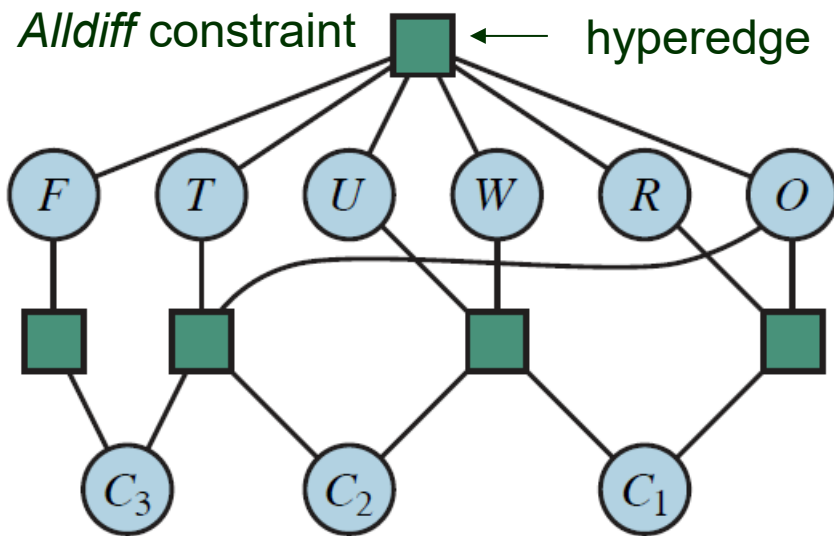
Constraint hypergraph

Cryptarithmic Puzzle

$$\begin{array}{r}
 T \quad W \quad O \\
 + \quad T \quad W \quad O \\
 \hline
 F \quad O \quad U \quad R
 \end{array}$$

Addition constraints on the four columns:

$$\begin{array}{l}
 O + O = R + 10 \cdot C_1 \\
 C_1 + W + W = U + 10 \cdot C_2 \\
 C_2 + T + T = O + 10 \cdot C_3 \\
 C_3 = F
 \end{array}
 \begin{array}{l}
 \boxed{C_1} \\
 \boxed{C_2} \\
 \boxed{C_3}
 \end{array}
 \begin{array}{l}
 \text{carry-overs into} \\
 \text{the tens, hundreds,} \\
 \text{and thousands,} \\
 \text{Respectively.}
 \end{array}$$



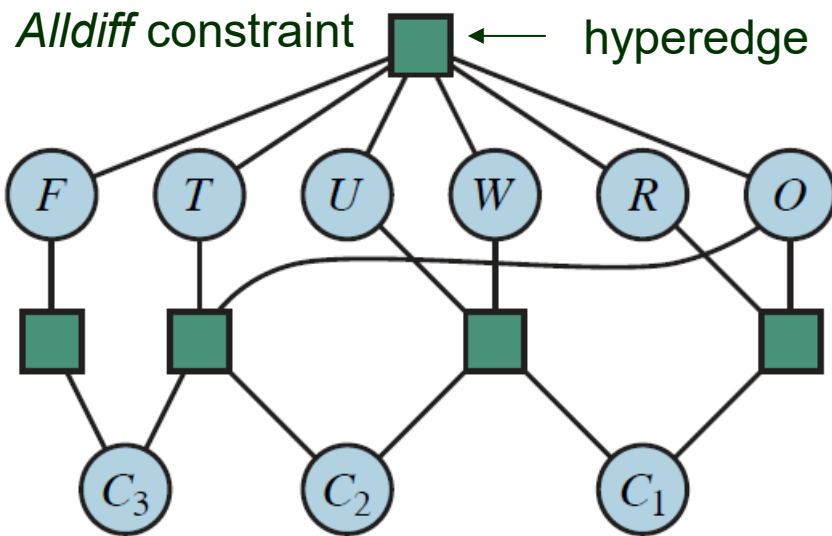
Constraint hypergraph

Cryptarithmic Puzzle

$$\begin{array}{r}
 T \quad W \quad O \\
 + \quad T \quad W \quad O \\
 \hline
 F \quad O \quad U \quad R
 \end{array}$$

Addition constraints on the four columns:

$$\begin{array}{l}
 O + O = R + 10 \cdot C_1 \\
 C_1 + W + W = U + 10 \cdot C_2 \\
 C_2 + T + T = O + 10 \cdot C_3 \\
 C_3 = F
 \end{array}
 \left. \begin{array}{l} C_1 \\ C_2 \\ C_3 \end{array} \right\} \begin{array}{l} \text{carry-overs into} \\ \text{the tens, hundreds,} \\ \text{and thousands,} \\ \text{Respectively.} \end{array}$$



A *hypergraph* generalizes a graph in that an edge (called an *hyperedge*) can connect more than two vertices.

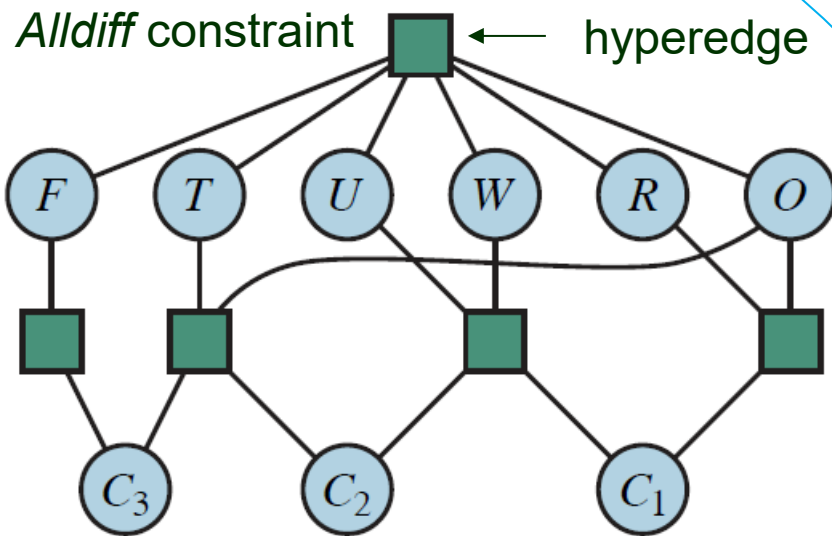
Constraint hypergraph

Cryptarithmic Puzzle

$$\begin{array}{r}
 T \quad W \quad O \\
 + \quad T \quad W \quad O \\
 \hline
 F \quad O \quad U \quad R
 \end{array}$$

Addition constraints on the four columns:

$$\begin{array}{l}
 O + O = R + 10 \cdot C_1 \\
 C_1 + W + W = U + 10 \cdot C_2 \\
 C_2 + T + T = O + 10 \cdot C_3 \\
 C_3 = F
 \end{array}
 \left. \begin{array}{l} C_1 \\ C_2 \\ C_3 \end{array} \right\} \begin{array}{l} \text{carry-overs into} \\ \text{the tens, hundreds,} \\ \text{and thousands,} \\ \text{Respectively.} \end{array}$$



A *hypergraph* generalizes a graph in that an edge (called an *hyperedge*) can connect more than two vertices.

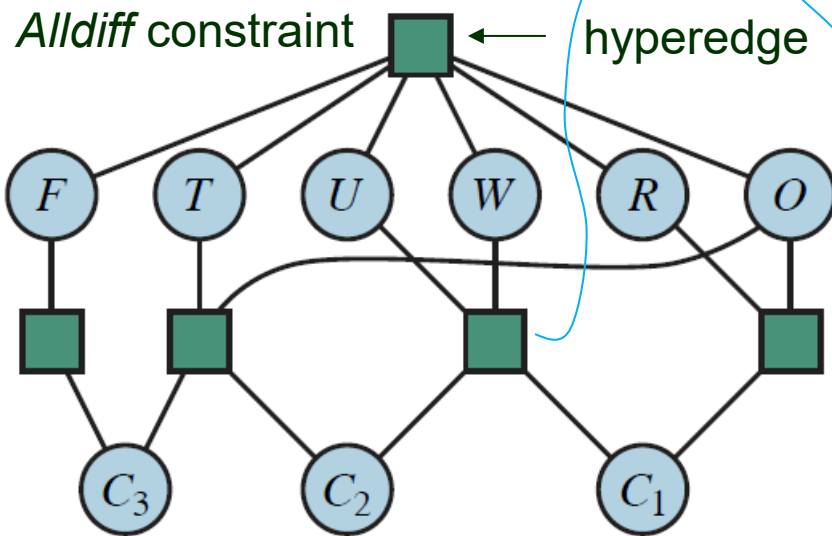
Constraint hypergraph

Cryptarithmic Puzzle

$$\begin{array}{r}
 T \quad W \quad O \\
 + \quad T \quad W \quad O \\
 \hline
 F \quad O \quad U \quad R
 \end{array}$$

Addition constraints on the four columns:

$$\begin{array}{l}
 O + O = R + 10 \cdot C_1 \\
 C_1 + W + W = U + 10 \cdot C_2 \\
 C_2 + T + T = O + 10 \cdot C_3 \\
 C_3 = F
 \end{array}
 \left. \begin{array}{l} C_1 \\ C_2 \\ C_3 \end{array} \right\} \begin{array}{l} \text{carry-overs into} \\ \text{the tens, hundreds,} \\ \text{and thousands,} \\ \text{Respectively.} \end{array}$$



A *hypergraph* generalizes a graph in that an edge (called an *hyperedge*) can connect more than two vertices.

Constraint hypergraph

Transformation to Binary Constraints

Any n -ary constraint can be reduced to a set of binary constraints by introducing auxiliary variables.

Transformation to Binary Constraints

Any n -ary constraint can be reduced to a set of binary constraints by introducing auxiliary variables.



Any CSP can be transformed into a binary CSP.

Transformation to Binary Constraints

Any n -ary constraint can be reduced to a set of binary constraints by introducing auxiliary variables.



Any CSP can be transformed into a binary CSP.

Advantages of a global constraint such as *Alldiff*:

- ◆ Easier and less error-prone to write.
- ◆ Allowing efficient special-purpose inference algorithms.

II. Constraint Propagation

Use constraints to reduce the number of legal values for a variable, which in turn reduces those for another variable, and so on.

II. Constraint Propagation

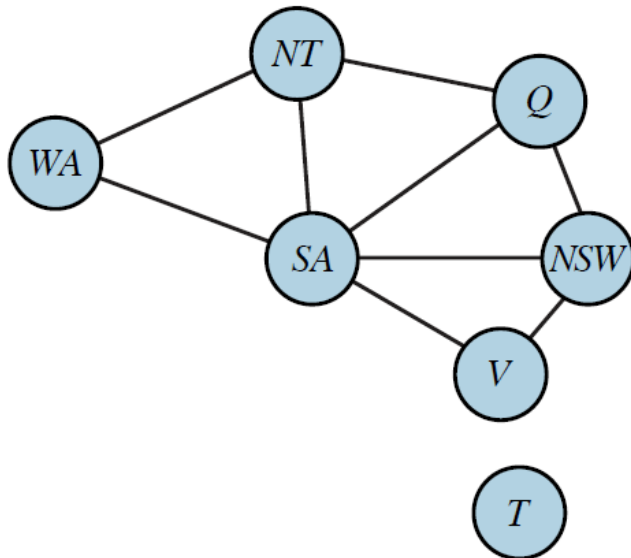
Use constraints to reduce the number of legal values for a variable, which in turn reduces those for another variable, and so on.

Enforcement of local consistency can cause inconsistent values to be eliminated throughout the constraint graph.

II. Constraint Propagation

Use constraints to reduce the number of legal values for a variable, which in turn reduces those for another variable, and so on.

Enforcement of local consistency can cause inconsistent values to be eliminated throughout the constraint graph.

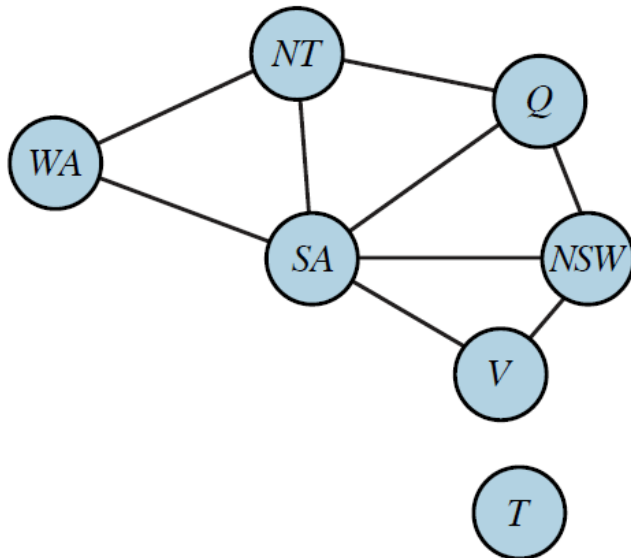


- Node consistency

II. Constraint Propagation

Use constraints to reduce the number of legal values for a variable, which in turn reduces those for another variable, and so on.

Enforcement of local consistency can cause inconsistent values to be eliminated throughout the constraint graph.



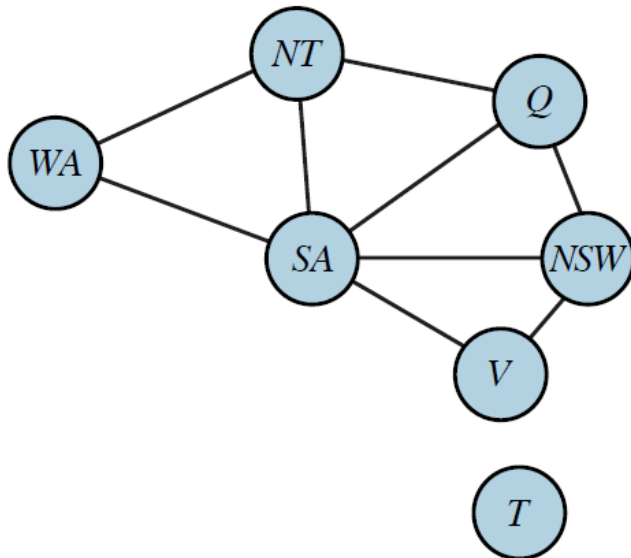
- Node consistency

Suppose South Australians dislike green.

II. Constraint Propagation

Use constraints to reduce the number of legal values for a variable, which in turn reduces those for another variable, and so on.

Enforcement of local consistency can cause inconsistent values to be eliminated throughout the constraint graph.



- Node consistency

Suppose South Australians dislike green.

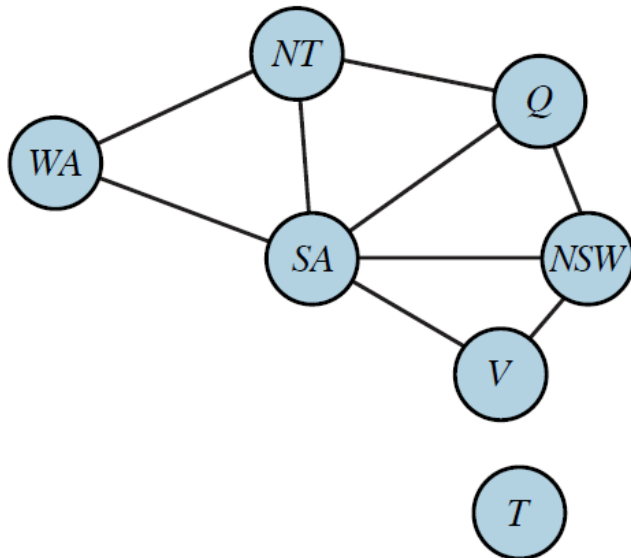


Domain for SA: {red, blue}

II. Constraint Propagation

Use constraints to reduce the number of legal values for a variable, which in turn reduces those for another variable, and so on.

Enforcement of local consistency can cause inconsistent values to be eliminated throughout the constraint graph.



- Node consistency

Suppose South Australians dislike green.



Domain for SA: {red, blue}

- ◆ Eliminate all the unary constraints by reducing the domains of the involved variables at the start.
- ◆ A variable is *node-consistent* if all the values in its domain satisfy its unary constraints.

Arc Consistency

A variable X_i is *arc-consistent* with respect to (w.r.t.) another variable X_j if for *every* value in D_i there exists a value in D_j that satisfies the binary constraint on the arc (i.e., edge (X_i, X_j)).



Arc Consistency

A variable X_i is *arc-consistent* with respect to (w.r.t.) another variable X_j if for *every* value in D_i there exists a value in D_j that satisfies the binary constraint on the arc (i.e., edge (X_i, X_j)).



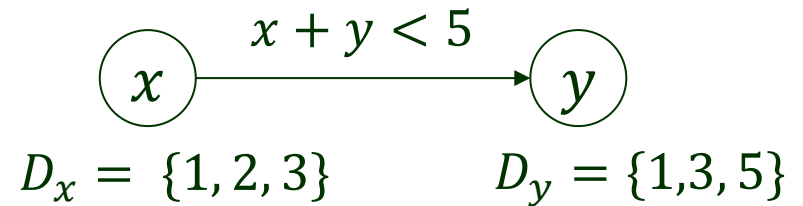
Non-commutative

Arc Consistency

A variable X_i is *arc-consistent* with respect to (w.r.t.) another variable X_j if for *every* value in D_i there exists a value in D_j that satisfies the binary constraint on the arc (i.e., edge (X_i, X_j)).



Non-commutative

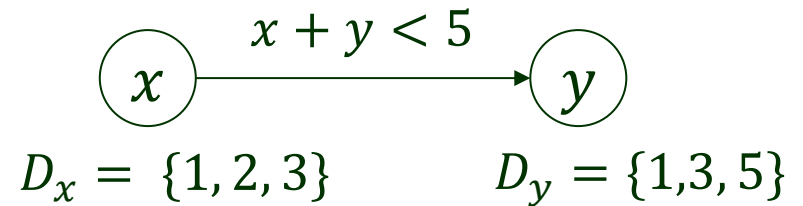


Arc Consistency

A variable X_i is *arc-consistent* with respect to (w.r.t.) another variable X_j if for *every* value in D_i there exists a value in D_j that satisfies the binary constraint on the arc (i.e., edge (X_i, X_j)).



Non-commutative



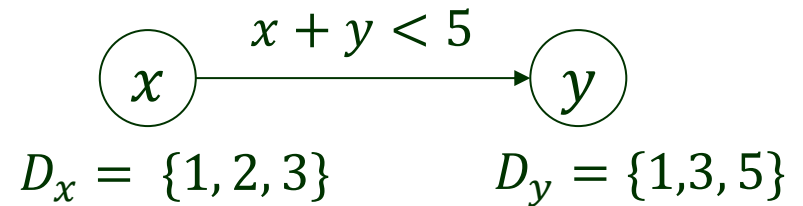
x is arc-consistent w.r.t. y ,
but not vice versa.

Arc Consistency

A variable X_i is *arc-consistent* with respect to (w.r.t.) another variable X_j if for *every* value in D_i there exists a value in D_j that satisfies the binary constraint on the arc (i.e., edge (X_i, X_j)).



Non-commutative



x is arc-consistent w.r.t. y ,
but not vice versa.

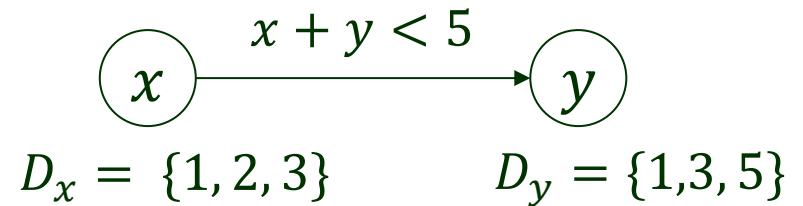
The variable is *arc-consistent* if it is so w.r.t every other variable that it shares a binary constraint with.

Arc Consistency

A variable X_i is *arc-consistent* with respect to (w.r.t.) another variable X_j if for *every* value in D_i there exists a value in D_j that satisfies the binary constraint on the arc (i.e., edge (X_i, X_j)).



Non-commutative



x is arc-consistent w.r.t. y ,
but not vice versa.

The variable is *arc-consistent* if it is so w.r.t every other variable that it shares a binary constraint with.

The constraint graph is *arc-consistent* if every variable is arc consistent with every other variable.

Application of Arc Consistency

Example 1

$$Y = X^2 \quad \text{where } X, Y \in \{0, 1, \dots, 9\}$$

Application of Arc Consistency

Example 1

$$Y = X^2 \quad \text{where } X, Y \in \{0, 1, \dots, 9\}$$

X arc-consistent w.r.t. Y

Application of Arc Consistency

Example 1

$$Y = X^2 \quad \text{where } X, Y \in \{0, 1, \dots, 9\}$$

X arc-consistent w.r.t. $Y \implies$ Reduced domain of X : $\{0, 1, 2, 3\}$

Application of Arc Consistency

Example 1

$$Y = X^2 \quad \text{where } X, Y \in \{0, 1, \dots, 9\}$$

X arc-consistent w.r.t. $Y \implies$ Reduced domain of X : $\{0, 1, 2, 3\}$

Y arc-consistent w.r.t. X

Application of Arc Consistency

Example 1

$$Y = X^2 \quad \text{where } X, Y \in \{0, 1, \dots, 9\}$$

X arc-consistent w.r.t. $Y \implies$ Reduced domain of X : $\{0, 1, 2, 3\}$

Y arc-consistent w.r.t. $X \implies$ Reduced domain of Y : $\{0, 1, 4, 9\}$

Application of Arc Consistency

Example 1

$$Y = X^2 \quad \text{where } X, Y \in \{0, 1, \dots, 9\}$$

X arc-consistent w.r.t. $Y \implies$ Reduced domain of X : $\{0, 1, 2, 3\}$

Y arc-consistent w.r.t. $X \implies$ Reduced domain of Y : $\{0, 1, 4, 9\}$

Example 2 Australia map-coloring

Application of Arc Consistency

Example 1

$$Y = X^2 \quad \text{where } X, Y \in \{0, 1, \dots, 9\}$$

X arc-consistent w.r.t. $Y \implies$ Reduced domain of X : $\{0, 1, 2, 3\}$

Y arc-consistent w.r.t. $X \implies$ Reduced domain of Y : $\{0, 1, 4, 9\}$

Example 2 Australia map-coloring

$SA \neq WA$

Application of Arc Consistency

Example 1

$$Y = X^2 \quad \text{where } X, Y \in \{0, 1, \dots, 9\}$$

X arc-consistent w.r.t. $Y \implies$ Reduced domain of X : $\{0, 1, 2, 3\}$

Y arc-consistent w.r.t. $X \implies$ Reduced domain of Y : $\{0, 1, 4, 9\}$

Example 2 Australia map-coloring

$SA \neq WA$

No effect on the domain $\{\text{red}, \text{green}, \text{blue}\}$ of each variable.

Arc-Consistency Algorithm AC-3

function AC-3(csp) **returns** false if an inconsistency is found and true otherwise

$queue \leftarrow$ a **queue** of arcs, initially all the arcs in csp
// (X_i, X_j) and (X_j, X_i) are both in $queue$.

while $queue$ is not empty **do**

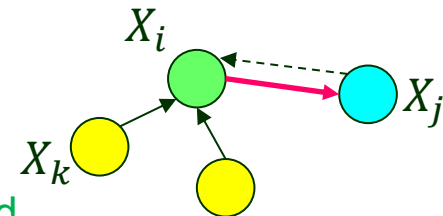
$(X_i, X_j) \leftarrow \text{POP}(queue)$

if REVISE(csp, X_i, X_j) **then** // domain of X_i has been reduced.

if size of $D_i = 0$ **then return** false

for each X_k **in** $X_i.\text{NEIGHBORS} - \{X_j\}$ **do** // propagate to other variables
add (X_k, X_i) to $queue$ // sharing a constraint with X_i

return true



function REVISE(csp, X_i, X_j) **returns** true iff we revise the domain of X_i

$revised \leftarrow false$

for each x **in** D_i **do**

if no value y in D_j allows (x, y) to satisfy the constraint between X_i and X_j **then**

delete x from D_i

$revised \leftarrow true$

return revised

Arc-Consistency Algorithm AC-3

function AC-3(*csp*) **returns** false if an inconsistency is found and true otherwise

queue $\leftarrow$ a **queue** of arcs, initially all the arcs in *csp*
// (X_i, X_j) and (X_j, X_i) are both in *queue*.

while *queue* is not empty **do**

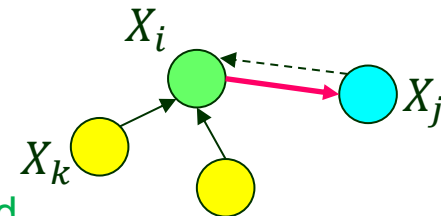
 (X_i, X_j) $\leftarrow$ POP(*queue*)

if REVISE(*csp*, X_i, X_j) **then** *// domain of X_i has been reduced.*

if size of $D_i = 0$ **then return** false

for each X_k **in** X_i .NEIGHBORS - $\{X_j\}$ **do** *// propagate to other variables*
 add (X_k, X_i) to *queue* *// sharing a constraint with X_i*

return true



function REVISE(*csp*, X_i, X_j) **returns** true iff we revise the domain of X_i

revised $\leftarrow$ false

for each x **in** D_i **do**

if no value y in D_j allows (x, y) to satisfy the constraint between X_i and X_j **then**

 delete x from D_i *// $O(d)$ time to check this condition*

revised $\leftarrow$ true

return *revised*

Domain size $\leq d$ & c binary constraints.

- Each arc inserted into the queue $\leq d$ times (X_i has $\leq d$ values to delete).
- Each checking takes $O(d^2)$ time.

Arc-Consistency Algorithm AC-3

function AC-3(*csp*) **returns** false if an inconsistency is found and true otherwise

queue $\leftarrow$ a **queue** of arcs, initially all the arcs in *csp*
// (X_i, X_j) and (X_j, X_i) are both in *queue*.

while *queue* is not empty **do**

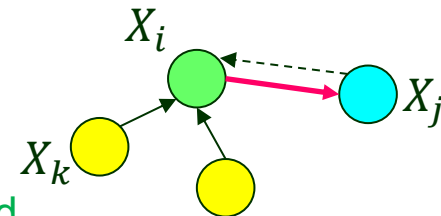
 (X_i, X_j) $\leftarrow$ POP(*queue*)

if REVISE(*csp*, X_i, X_j) **then** *// domain of X_i has been reduced.*

if size of $D_i = 0$ **then return** false

for each X_k **in** X_i .NEIGHBORS - $\{X_j\}$ **do** *// propagate to other variables*
 add (X_k, X_i) to *queue* *// sharing a constraint with X_i*

return true



function REVISE(*csp*, X_i, X_j) **returns** true iff we revise the domain of X_i

revised $\leftarrow$ false

for each x **in** D_i **do**

if no value y in D_j allows (x, y) to satisfy the constraint between X_i and X_j **then**

 delete x from D_i *// $O(d)$ time to check this condition*

revised $\leftarrow$ true

return *revised*

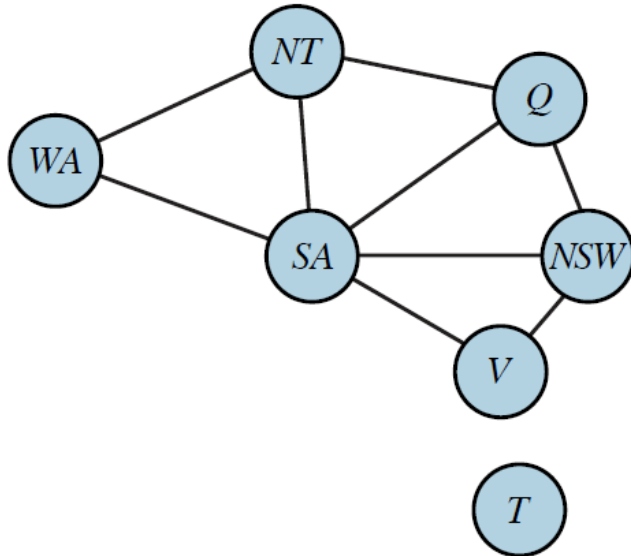
Domain size $\leq d$ & c binary constraints.

- Each arc inserted into the queue $\leq d$ times (X_i has $\leq d$ values to delete).
- Each checking takes $O(d^2)$ time.

$O(cd^3)$

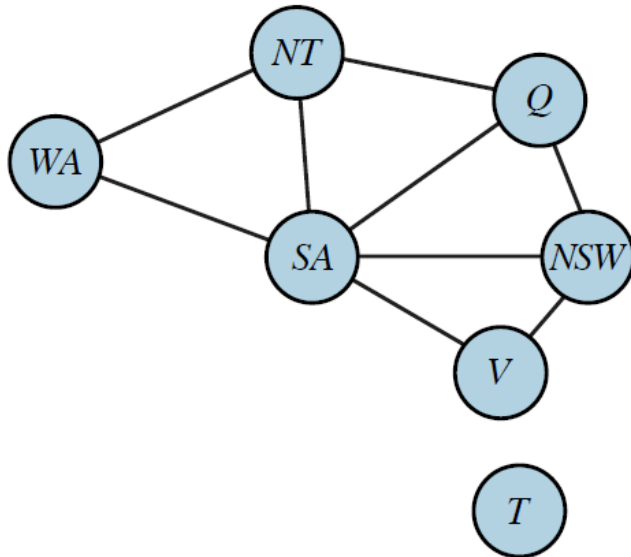
Path Consistency

Arc consistency reduces the domains.



Path Consistency

Arc consistency reduces the domains.

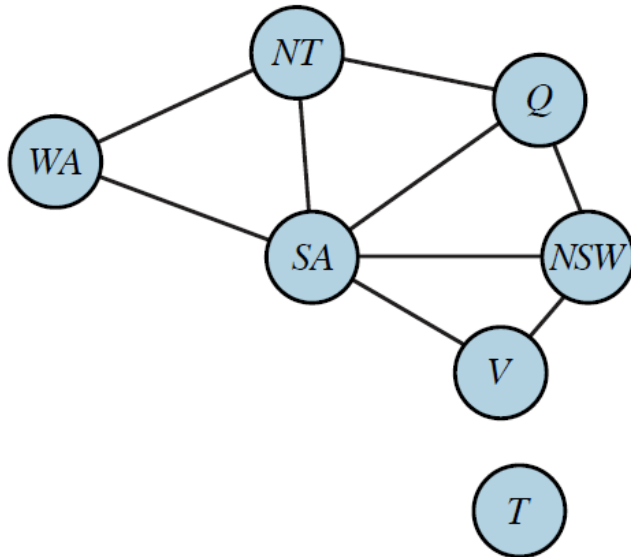


No solution if using two colors
even though every constraint
can be satisfied individually.

Path Consistency

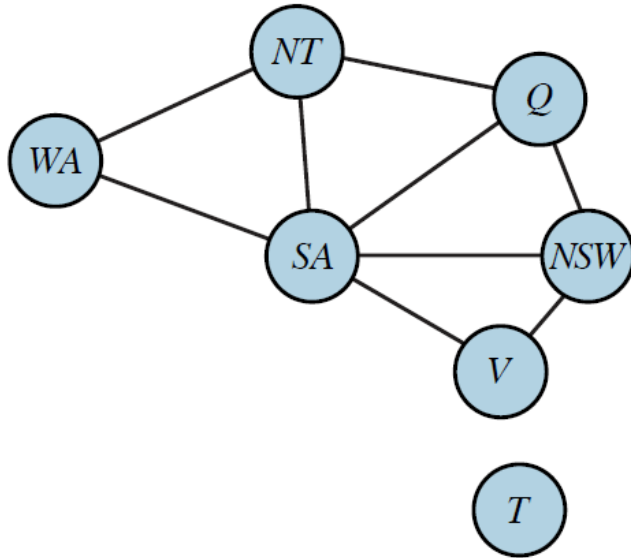
Arc consistency reduces the domains.

Path consistency checks implicit constraints inferable across triples of variables along a path.



No solution if using two colors even though every constraint can be satisfied individually.

Path Consistency

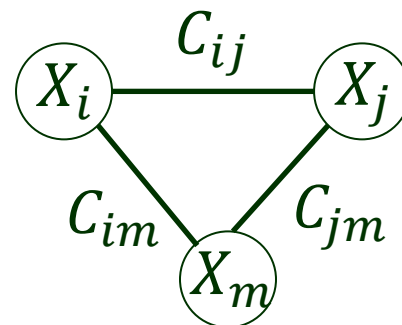


No solution if using two colors even though every constraint can be satisfied individually.

Arc consistency reduces the domains.

Path consistency checks implicit constraints inferable across triples of variables along a path.

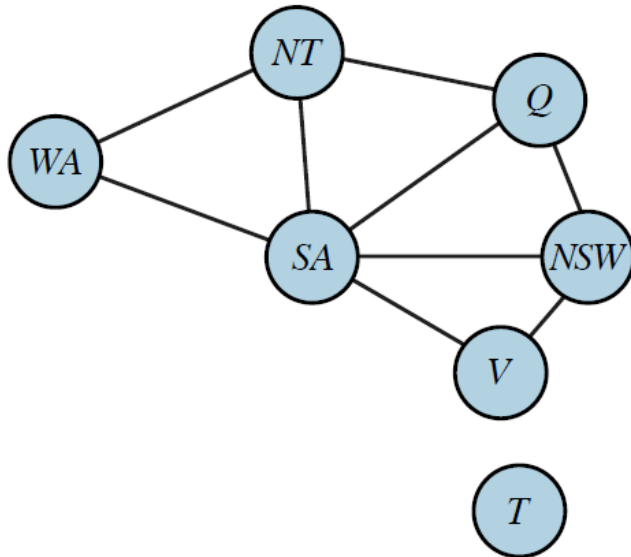
$\{X_i, X_j\}$ is *path-consistent* w.r.t. X_m if for every assignment to X_i, X_j consistent with their constraint C_{ij} (if exists), there exists an assignment to X_m that satisfies the constraints C_{im} and C_{jm} between them and X_m .



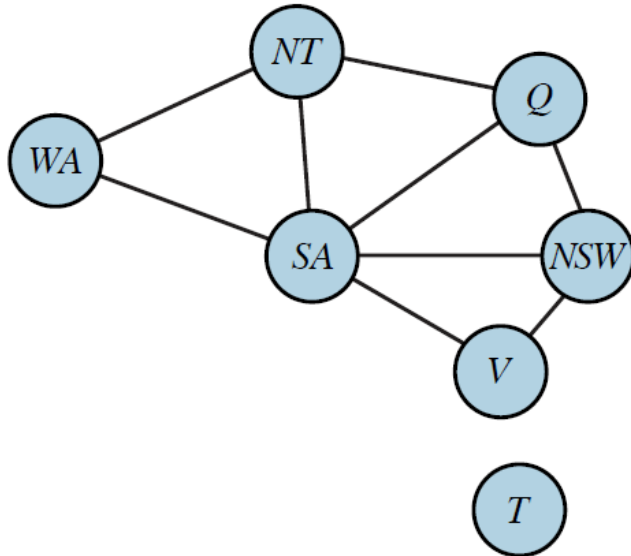
Application of Path Consistency

Make $\{WA, SA\}$ path-consistent w.r.t. NT .

Assume two colors only: *red*, *blue*.



Application of Path Consistency



Make $\{WA, SA\}$ path-consistent w.r.t. NT .

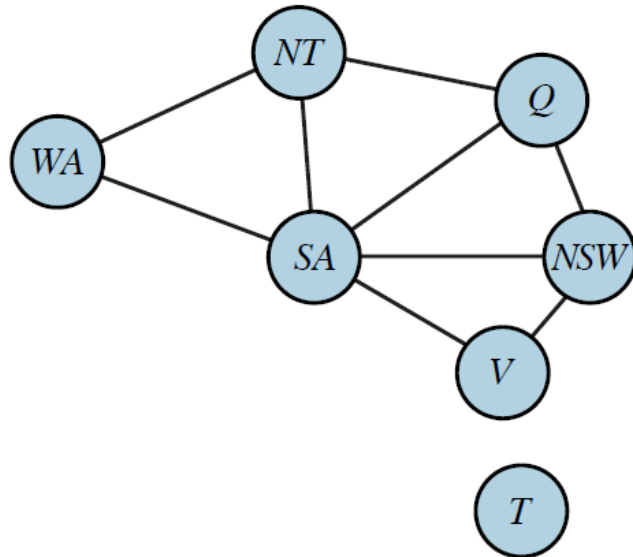
Assume two colors only: *red*, *blue*.

Only two possible assignments:

$\{WA = \text{red}, SA = \text{blue}\}$

$\{WA = \text{blue}, SA = \text{red}\}$

Application of Path Consistency



Make $\{WA, SA\}$ path-consistent w.r.t. NT .

Assume two colors only: *red*, *blue*.

Only two possible assignments:

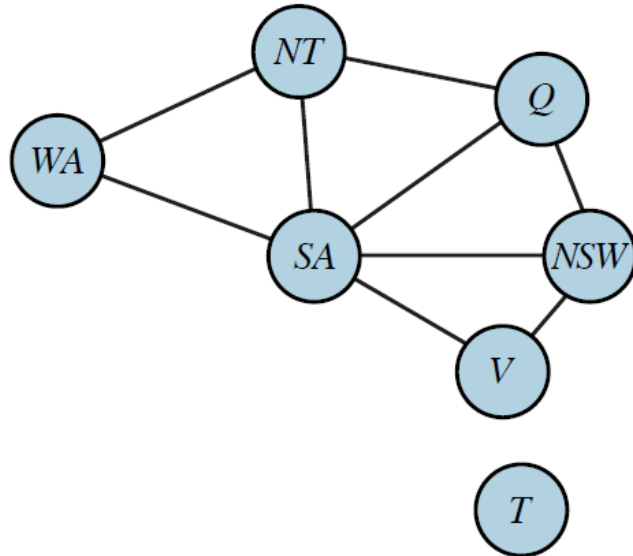
$\{WA = \text{red}, SA = \text{blue}\}$

$\{WA = \text{blue}, SA = \text{red}\}$



No assignment for NT in either case!

Application of Path Consistency



Make $\{WA, SA\}$ path-consistent w.r.t. NT .

Assume two colors only: *red*, *blue*.

Only two possible assignments:

$\{WA = \text{red}, SA = \text{blue}\}$

$\{WA = \text{blue}, SA = \text{red}\}$

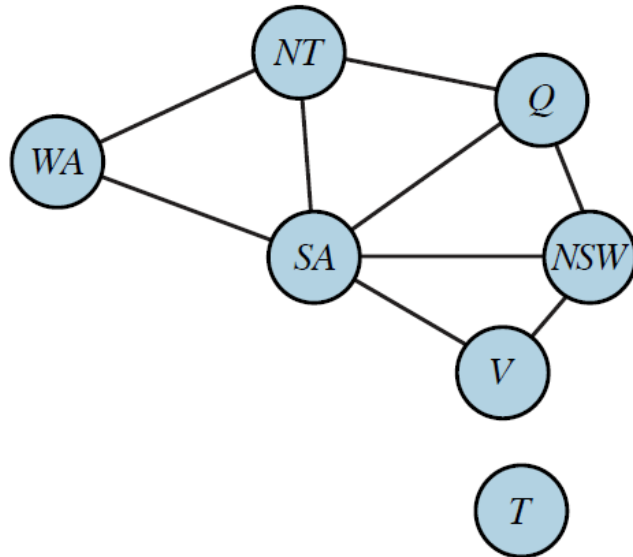


No assignment for NT in either case!



Eliminate both assignments to WA and SA .

Application of Path Consistency



Make $\{WA, SA\}$ path-consistent w.r.t. NT .

Assume two colors only: *red*, *blue*.

Only two possible assignments:

$\{WA = \text{red}, SA = \text{blue}\}$

$\{WA = \text{blue}, SA = \text{red}\}$



No assignment for NT in either case!



Eliminate both assignments to WA and SA .



No solution to the problem.

Bounds Propagation

Two flights F_1 and F_2 have domains:

$$D_1 = [0,165] \text{ and } D_2 = [0,385]$$

max capacity

Bounds Propagation

Two flights F_1 and F_2 have domains:

$$D_1 = [0,165] \text{ and } D_2 = [0,385]$$

max capacity

Constraint: the two flights together carry 420 people.

Bounds Propagation

Two flights F_1 and F_2 have domains:

$$D_1 = [0,165] \text{ and } D_2 = [0,385]$$

↓
max capacity

Constraint: the two flights together carry 420 people.

↓ domain reduction

$$D_1 = [420 - 385, 165] = [35, 165]$$

$$D_2 = [420 - 165, 385] = [255, 385]$$

Bounds Propagation

Two flights F_1 and F_2 have domains:

$$D_1 = [0,165] \text{ and } D_2 = [0,385]$$

|
max capacity

Constraint: the two flights together carry 420 people.

↓ domain reduction

$$D_1 = [420 - 385, 165] = [35, 165]$$

$$D_2 = [420 - 165, 385] = [255, 385]$$

Bounds-consistent if for any two variables X, Y , whether X takes on its lower- or upper-bound values, there exists some value of Y that satisfies the constraint between X and Y .

Bounds Propagation

Two flights F_1 and F_2 have domains:

$$D_1 = [0,165] \quad \text{and} \quad D_2 = [0,385]$$

↓
max capacity

Constraint: the two flights together carry 420 people.

↓ domain reduction

$$D_1 = [420 - 385, 165] = [35, 165]$$

$$D_2 = [420 - 165, 385] = [255, 385]$$

Bounds-consistent if for any two variables X, Y , whether X takes on its lower- or upper-bound values, there exists some value of Y that satisfies the constraint between X and Y .

* By the above definition, for some value of X between its lower and upper bounds, there may not exist a value of Y to meet the constraint (e.g., a nonlinear constraint). e.g., for $X \in [0, \pi], Y \in [0, \frac{1}{2}]$ the constraint $Y = \sin X$ is not satisfied at $X = \pi/2$.

Sudoku

Fill the digits 1 to 9 in a 9×9 grid such that **no digit appears twice** in any row, column, or 3×3 box.

	1	2	3	4	5	6	7	8	9
A			3		2		6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7								8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1		3		

Sudoku

Fill the digits 1 to 9 in a 9×9 grid such that **no digit appears twice** in any row, column, or 3×3 box.

	1	2	3	4	5	6	7	8	9
A			3		2		6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7								8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1		3		

	1	2	3	4	5	6	7	8	9
A	4	8	3	9	2	1	6	5	7
B	9	6	7	3	4	5	8	2	1
C	2	5	1	8	7	6	4	9	3
D	5	4	8	1	3	2	9	7	6
E	7	2	9	5	6	4	1	3	8
F	1	3	6	7	9	8	2	4	5
G	3	7	2	6	8	9	5	1	4
H	8	1	4	2	5	3	7	6	9
I	6	9	5	4	1	7	3	8	2

Sudoku as a CSP

	1	2	3	4	5	6	7	8	9
A			3		2		6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7								8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1		3		

Variables: $A1, \dots, A9, B1, \dots, I9$

Domain: $D = \{1, 2, \dots, 9\}$

Sudoku as a CSP

	1	2	3	4	5	6	7	8	9
A			3		2		6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7								8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1		3		

Variables: $A1, \dots, A9, B1, \dots, I9$

Domain: $D = \{1, 2, \dots, 9\}$

27 *Alldiff* constraints:

Sudoku as a CSP

	1	2	3	4	5	6	7	8	9
A			3		2		6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7								8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1		3		

Variables: $A1, \dots, A9, B1, \dots, I9$

Domain: $D = \{1, 2, \dots, 9\}$

27 *Alldiff* constraints:

Alldiff($A1, A2, A3, A4, A5, A6, A7, A8, A9$)

$\vdots$

Alldiff($I1, I2, I3, I4, I5, I6, I7, I8, I9$)

Sudoku as a CSP

	1	2	3	4	5	6	7	8	9
A			3		2		6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7								8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1		3		

Variables: $A1, \dots, A9, B1, \dots, I9$

Domain: $D = \{1, 2, \dots, 9\}$

27 *Alldiff* constraints:

Alldiff($A1, A2, A3, A4, A5, A6, A7, A8, A9$)

⋮

Alldiff($I1, I2, I3, I4, I5, I6, I7, I8, I9$)

Alldiff($A1, B1, C1, D1, E1, F1, G1, H1, I1$)

⋮

Alldiff($A9, B9, C9, D9, E9, F9, G9, H9, I9$)

Sudoku as a CSP

	1	2	3	4	5	6	7	8	9
A			3		2		6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7								8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1		3		

Variables: $A1, \dots, A9, B1, \dots, I9$

Domain: $D = \{1, 2, \dots, 9\}$

27 *Alldiff* constraints:

Alldiff($A1, A2, A3, A4, A5, A6, A7, A8, A9$)

⋮

Alldiff($I1, I2, I3, I4, I5, I6, I7, I8, I9$)

Alldiff($A1, B1, C1, D1, E1, F1, G1, H1, I1$)

⋮

Alldiff($A9, B9, C9, D9, E9, F9, G9, H9, I9$)

Alldiff($A1, A2, A3, B1, B2, B3, C1, C2, C3$)

⋮

Alldiff($G7, G8, G9, H7, H8, H9, I7, I8, I9$)

Sudoku as a CSP

	1	2	3	4	5	6	7	8	9
A			3		2		6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7								8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1		3		

Variables: $A1, \dots, A9, B1, \dots, I9$

Domain: $D = \{1, 2, \dots, 9\}$

27 *Alldiff* constraints:

Alldiff($A1, A2, A3, A4, A5, A6, A7, A8, A9$)

⋮

Alldiff($I1, I2, I3, I4, I5, I6, I7, I8, I9$)

Alldiff($A1, B1, C1, D1, E1, F1, G1, H1, I1$)

⋮

Alldiff($A9, B9, C9, D9, E9, F9, G9, H9, I9$)

Alldiff($A1, A2, A3, B1, B2, B3, C1, C2, C3$)

⋮

Alldiff($G7, G8, G9, H7, H8, H9, I7, I8, I9$)

A CSP solver can handle thousands of puzzles per second!

Sudoku as a CSP

	1	2	3	4	5	6	7	8	9
A			3		2		6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7								8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1		3		

Variables: $A1, \dots, A9, B1, \dots, I9$

Domain: $D = \{1, 2, \dots, 9\}$

27 *Alldiff* constraints:

Alldiff($A1, A2, A3, A4, A5, A6, A7, A8, A9$)

⋮

Alldiff($I1, I2, I3, I4, I5, I6, I7, I8, I9$)

Alldiff($A1, B1, C1, D1, E1, F1, G1, H1, I1$)

⋮

Alldiff($A9, B9, C9, D9, E9, F9, G9, H9, I9$)

Alldiff($A1, A2, A3, B1, B2, B3, C1, C2, C3$)

⋮

Alldiff($G7, G8, G9, H7, H8, H9, I7, I8, I9$)

A CSP solver can handle thousands of puzzles per second!
Only the simplest ones can be solved by AC-3.

Example of CSP Solution

	1	2	3	4	5	6	7	8	9
A			3		2		6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7								8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1		3		

Example of CSP Solution

	1	2	3	4	5	6	7	8	9
A			3		2		6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7					?			8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1		3		

Consider E6:

Example of CSP Solution

	1	2	3	4	5	6	7	8	9
A			3		2		6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7					?			8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1		3		

Consider E6:

$$D_{E6} \leftarrow \{1, 2, \dots, 9\}$$

Example of CSP Solution

	1	2	3	4	5	6	7	8	9
A			3		2		6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7					?			8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1		3		

Consider E6:

$$D_{E6} \leftarrow \{1, 2, \dots, 9\}$$

↓ constraints in the box

$$\begin{aligned} D_{E6} &\leftarrow D_{E6} \setminus \{1, 2, 7, 8\} \\ &= \{3, 4, 5, 6, 9\} \end{aligned}$$

Example of CSP Solution

	1	2	3	4	5	6	7	8	9
A			3		2		6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7					?			8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1		3		

Consider E6:

$$D_{E6} \leftarrow \{1, 2, \dots, 9\}$$

↓ constraints in the box

$$\begin{aligned} D_{E6} &\leftarrow D_{E6} \setminus \{1, 2, 7, 8\} \\ &= \{3, 4, 5, 6, 9\} \end{aligned}$$

↓ constraints in the column

$$\begin{aligned} D_{E6} &\leftarrow D_{E6} \setminus \{2, 3, 5, 6, 8, 9\} \\ &= \{4\} \end{aligned}$$

Example of CSP Solution

	1	2	3	4	5	6	7	8	9
A			3		2		6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7					4			8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1		3		

Consider E6:

$$D_{E6} \leftarrow \{1, 2, \dots, 9\}$$

↓ constraints in the box

$$\begin{aligned} D_{E6} &\leftarrow D_{E6} \setminus \{1, 2, 7, 8\} \\ &= \{3, 4, 5, 6, 9\} \end{aligned}$$

↓ constraints in the column

$$\begin{aligned} D_{E6} &\leftarrow D_{E6} \setminus \{2, 3, 5, 6, 8, 9\} \\ &= \{4\} \end{aligned}$$

Example of CSP Solution

	1	2	3	4	5	6	7	8	9
A			3		2		6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7					4			8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1	?	3		

Consider I6:

Consider E6:

$$D_{E6} \leftarrow \{1, 2, \dots, 9\}$$

↓ constraints in the box

$$\begin{aligned} D_{E6} &\leftarrow D_{E6} \setminus \{1, 2, 7, 8\} \\ &= \{3, 4, 5, 6, 9\} \end{aligned}$$

↓ constraints in the column

$$\begin{aligned} D_{E6} &\leftarrow D_{E6} \setminus \{2, 3, 5, 6, 8, 9\} \\ &= \{4\} \end{aligned}$$

Example of CSP Solution

	1	2	3	4	5	6	7	8	9
A			3		2		6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7					4			8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1	?	3		

Consider I6: $D_{I6} \leftarrow \{1, 2, \dots, 9\}$

Consider E6:

$$D_{E6} \leftarrow \{1, 2, \dots, 9\}$$

↓ constraints in the box

$$D_{E6} \leftarrow D_{E6} \setminus \{1, 2, 7, 8\}$$

$$= \{3, 4, 5, 6, 9\}$$

↓ constraints in the column

$$D_{E6} \leftarrow D_{E6} \setminus \{2, 3, 5, 6, 8, 9\}$$

$$= \{4\}$$

Example of CSP Solution

	1	2	3	4	5	6	7	8	9
A			3		2		6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7					4			8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1	?	3		

Consider E6:

$$D_{E6} \leftarrow \{1, 2, \dots, 9\}$$

↓ constraints in the box

$$D_{E6} \leftarrow D_{E6} \setminus \{1, 2, 7, 8\}$$

$$= \{3, 4, 5, 6, 9\}$$

↓ constraints in the column

$$D_{E6} \leftarrow D_{E6} \setminus \{2, 3, 5, 6, 8, 9\}$$

$$= \{4\}$$

Consider I6:

$$D_{I6} \leftarrow \{1, 2, \dots, 9\}$$

↓ constraints in the column

$$D_{I6} \leftarrow D_{I6} \setminus \{2, 3, 4, 5, 6, 8, 9\} = \{1, 7\}$$

Example of CSP Solution

	1	2	3	4	5	6	7	8	9
A			3		2		6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7					4			8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1	?	3		

Consider E6:

$$D_{E6} \leftarrow \{1, 2, \dots, 9\}$$

↓ constraints in the box

$$D_{E6} \leftarrow D_{E6} \setminus \{1, 2, 7, 8\}$$

$$= \{3, 4, 5, 6, 9\}$$

↓ constraints in the column

$$D_{E6} \leftarrow D_{E6} \setminus \{2, 3, 5, 6, 8, 9\}$$

$$= \{4\}$$

Consider I6:

$$D_{I6} \leftarrow \{1, 2, \dots, 9\}$$

↓ constraints in the column

$$D_{I6} \leftarrow D_{I6} \setminus \{2, 3, 4, 5, 6, 8, 9\} = \{1, 7\}$$

↓ constraints in the box

$$D_{I6} \leftarrow D_{I6} \setminus \{1, 2, 3, 6, 9\} = \{7\}$$

Example of CSP Solution

	1	2	3	4	5	6	7	8	9
A			3		2		6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7					4			8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1	7	3		

Consider E6:

$$D_{E6} \leftarrow \{1, 2, \dots, 9\}$$

↓ constraints in the box

$$D_{E6} \leftarrow D_{E6} \setminus \{1, 2, 7, 8\}$$

$$= \{3, 4, 5, 6, 9\}$$

↓ constraints in the column

$$D_{E6} \leftarrow D_{E6} \setminus \{2, 3, 5, 6, 8, 9\}$$

$$= \{4\}$$

Consider I6:

$$D_{I6} \leftarrow \{1, 2, \dots, 9\}$$

↓ constraints in the column

$$D_{I6} \leftarrow D_{I6} \setminus \{2, 3, 4, 5, 6, 8, 9\} = \{1, 7\}$$

↓ constraints in the box

$$D_{I6} \leftarrow D_{I6} \setminus \{1, 2, 3, 6, 9\} = \{7\}$$

Example of CSP Solution

	1	2	3	4	5	6	7	8	9
A			3		2	?	6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7					4			8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1	7	3		

Consider E6:

$$D_{E6} \leftarrow \{1, 2, \dots, 9\}$$

↓ constraints in the box

$$D_{E6} \leftarrow D_{E6} \setminus \{1, 2, 7, 8\}$$

$$= \{3, 4, 5, 6, 9\}$$

↓ constraints in the column

$$D_{E6} \leftarrow D_{E6} \setminus \{2, 3, 5, 6, 8, 9\}$$

$$= \{4\}$$

Consider A6:

Consider I6: $D_{I6} \leftarrow \{1, 2, \dots, 9\}$

↓ constraints in the column

$$D_{I6} \leftarrow D_{I6} \setminus \{2, 3, 4, 5, 6, 8, 9\} = \{1, 7\}$$

↓ constraints in the box

$$D_{I6} \leftarrow D_{I6} \setminus \{1, 2, 3, 6, 9\} = \{7\}$$

Example of CSP Solution

	1	2	3	4	5	6	7	8	9
A			3		2	?	6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7					4			8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1	7	3		

Consider I6: $D_{I6} \leftarrow \{1, 2, \dots, 9\}$
 $\Downarrow$ constraints in the column

$$D_{I6} \leftarrow D_{I6} \setminus \{2, 3, 4, 5, 6, 8, 9\} = \{1, 7\}$$

$$\Downarrow \text{constraints in the box}$$

$$D_{I6} \leftarrow D_{I6} \setminus \{1, 2, 3, 6, 9\} = \{7\}$$

Consider E6:

$$D_{E6} \leftarrow \{1, 2, \dots, 9\}$$

$\Downarrow$ constraints in the box

$$D_{E6} \leftarrow D_{E6} \setminus \{1, 2, 7, 8\}$$

$$= \{3, 4, 5, 6, 9\}$$

$\Downarrow$ constraints in the column

$$D_{E6} \leftarrow D_{E6} \setminus \{2, 3, 5, 6, 8, 9\}$$

$$= \{4\}$$

Consider A6:

$$D_{A6} \leftarrow \{1, 2, \dots, 9\}$$

Example of CSP Solution

	1	2	3	4	5	6	7	8	9
A			3		2	?	6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7					4			8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1	7	3		

Consider I6: $D_{I6} \leftarrow \{1, 2, \dots, 9\}$
 $\Downarrow$ constraints in the column
 $D_{I6} \leftarrow D_{I6} \setminus \{2, 3, 4, 5, 6, 8, 9\} = \{1, 7\}$
 $\Downarrow$ constraints in the box
 $D_{I6} \leftarrow D_{I6} \setminus \{1, 2, 3, 6, 9\} = \{7\}$

Consider E6:

$D_{E6} \leftarrow \{1, 2, \dots, 9\}$
 $\Downarrow$ constraints in the box
 $D_{E6} \leftarrow D_{E6} \setminus \{1, 2, 7, 8\}$
 $= \{3, 4, 5, 6, 9\}$
 $\Downarrow$ constraints in the column
 $D_{E6} \leftarrow D_{E6} \setminus \{2, 3, 5, 6, 8, 9\}$
 $= \{4\}$

Consider A6:

$D_{A6} \leftarrow \{1, 2, \dots, 9\}$
 $\Downarrow$ constraints in the column
 $D_{A6} \leftarrow \{1\}$

Example of CSP Solution

	1	2	3	4	5	6	7	8	9
A			3		2	1	6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7					4			8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1	7	3		

Consider I6: $D_{I6} \leftarrow \{1, 2, \dots, 9\}$
 $\Downarrow$ constraints in the column
 $D_{I6} \leftarrow D_{I6} \setminus \{2, 3, 4, 5, 6, 8, 9\} = \{1, 7\}$
 $\Downarrow$ constraints in the box
 $D_{I6} \leftarrow D_{I6} \setminus \{1, 2, 3, 6, 9\} = \{7\}$

Consider E6:

$D_{E6} \leftarrow \{1, 2, \dots, 9\}$
 $\Downarrow$ constraints in the box
 $D_{E6} \leftarrow D_{E6} \setminus \{1, 2, 7, 8\}$
 $= \{3, 4, 5, 6, 9\}$
 $\Downarrow$ constraints in the column
 $D_{E6} \leftarrow D_{E6} \setminus \{2, 3, 5, 6, 8, 9\}$
 $= \{4\}$

Consider A6:

$D_{A6} \leftarrow \{1, 2, \dots, 9\}$
 $\Downarrow$ constraints in the column
 $D_{A6} \leftarrow \{1\}$